

The Complete Treatise on the General Quartic

Soumadeep Ghosh

Kolkata, India

Abstract

The general quartic equation represents the highest-degree generic polynomial equation solvable by radicals. This paper presents a rigorous and interdisciplinary treatment of the quartic equation, beginning with its historical origins and culminating in modern computational, geometric, statistical, economic, and strategic interpretations. Ferrari's method is derived carefully from first principles, including the construction of the resolvent cubic and the complete algebraic factorization procedure.

The quartic is then examined from the perspectives of algorithmic complexity, nonlinear optimization, artificial intelligence, financial instability, strategic equilibrium analysis, and geopolitical systems. The paper argues that the quartic is not merely an algebraic object but a universal non-linear structural archetype governing systems with multiple equilibria, bifurcations, and instability transitions.

The manuscript uses only standard L^AT_EX packages and provides mathematically rigorous derivations, tables, algorithms, and PGF/TikZ vector graphics suitable for academic publication.

The paper ends with "The End"

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1 Introduction

The general quartic takes the form

$$Ax^4 + Bx^3 + Cx^2 + Dx + E = 0, A \neq 0.$$

After dividing by $A \neq 0$, the general monic quartic takes the form

$$x^4 + ax^3 + bx^2 + cx + d = 0.$$

The quartic occupies a uniquely important position in algebra. Linear, quadratic, cubic, and quartic equations admit general radical solutions. Historically, the quartic solution emerged during the sixteenth-century Italian algebraic revolution through the work of Lodovico Ferrari, extending earlier methods developed by Tartaglia and Cardano for the cubic equation.

Modern mathematics interprets the quartic simultaneously as:

- an algebraic object,
- a geometric structure,
- a dynamical system,
- a computational problem,
- an optimization landscape,
- and a nonlinear equilibrium mechanism.

2 Historical Background

The development of quartic methods transformed symbolic mathematics.

Mathematician	Contribution
Scipione del Ferro	Depressed cubic solution
Niccol'ò Tartaglia	Independent cubic methods
Gerolamo Cardano	Publication of cubic formula
Lodovico Ferrari	General quartic solution

Table 1: Historical development of polynomial solvability

Ferrari's method established several foundational principles of modern mathematics:

1. algebraic reduction,
2. recursive decomposition,
3. auxiliary-variable construction,
4. structural factorization.

These ideas later generalized into symbolic computation, algorithm design, and modern algebra.

3 Reduction to the Depressed Quartic

3.1 Elimination of the Cubic Term

Consider

$$x^4 + ax^3 + bx^2 + cx + d = 0.$$

Introduce the substitution

$$x = y - \frac{a}{4}.$$

Substituting and simplifying yields

$$y^4 + py^2 + qy + r = 0,$$

where

$$p = b - \frac{3a^2}{8},$$

$$q = c - \frac{ab}{2} + \frac{a^3}{8},$$

$$r = d - \frac{ac}{4} + \frac{a^2b}{16} - \frac{3a^4}{256}.$$

The transformed equation is called the depressed quartic.

3.2 Interpretation

The Tschirnhaus transformation removes asymmetry from the polynomial and normalizes the coordinate system.

Analogous normalization procedures appear in:

- gauge fixing in physics,
- centering transformations in statistics,
- detrending in econometrics,
- coordinate normalization in machine learning.

4 Ferrari's Method

4.1 Core Construction

Start from the depressed quartic:

$$y^4 + py^2 + qy + r = 0.$$

Rewrite it as

$$(y^2 + \alpha)^2 - [(2\alpha - p)y^2 - qy + (\alpha^2 - r)] = 0.$$

We now choose α so that the bracketed expression becomes a perfect square.

Assume

$$(2\alpha - p)y^2 - qy + (\alpha^2 - r) = (my - n)^2.$$

Expanding the right-hand side gives

$$m^2y^2 - 2mny + n^2.$$

Matching coefficients yields

$$m^2 = 2\alpha - p,$$

$$2mn = q,$$

$$n^2 = \alpha^2 - r.$$

Eliminating m and n :

$$4(\alpha^2 - r)(2\alpha - p) = q^2.$$

Expanding:

$$8\alpha^3 - 4p\alpha^2 - 8r\alpha + 4pr - q^2 = 0.$$

Dividing by 4 gives the resolvent cubic:

$$2\alpha^3 - p\alpha^2 - 2r\alpha + pr - \frac{q^2}{4} = 0.$$

5 Factorization of the Quartic

Let α be a solution of the resolvent cubic.

Define

$$m = \sqrt{2\alpha - p}.$$

Assuming $m \neq 0$, define

$$n = \frac{q}{2m}.$$

Then

$$(y^2 + \alpha)^2 - (my - n)^2 = 0.$$

Using the difference-of-squares identity:

$$A^2 - B^2 = (A - B)(A + B),$$

we obtain

$$(y^2 - my + \alpha + n)(y^2 + my + \alpha - n) = 0.$$

Thus the quartic factors into two quadratic equations:

$$y^2 - my + \alpha + n = 0,$$

$$y^2 + my + \alpha - n = 0.$$

Applying the quadratic formula:

$$y = \frac{m \pm \sqrt{m^2 - 4(\alpha + n)}}{2},$$

and

$$y = \frac{-m \pm \sqrt{m^2 - 4(\alpha - n)}}{2}.$$

Finally,

$$x = y - \frac{a}{4}.$$

6 Proof of Solvability

Theorem 1. *Every quartic polynomial over a field of characteristic not equal to 2 is solvable by radicals.*

Proof. A quartic equation can be reduced to a depressed quartic by a linear substitution.

Ferrari's method reduces the depressed quartic to:

1. one cubic equation,
2. two quadratic equations.

Since cubic equations are solvable by radicals and quadratic equations are solvable by radicals, the quartic itself is solvable by radicals. □

7 The Ferrari Algorithm

7.1 Algorithmic Procedure

Given

$$x^4 + ax^3 + bx^2 + cx + d = 0,$$

perform the following steps.

1. Compute

$$p = b - \frac{3a^2}{8},$$

$$q = c - \frac{ab}{2} + \frac{a^3}{8},$$

$$r = d - \frac{ac}{4} + \frac{a^2b}{16} - \frac{3a^4}{256}.$$

2. Solve the resolvent cubic:

$$2\alpha^3 - p\alpha^2 - 2r\alpha + pr - \frac{q^2}{4} = 0.$$

3. Choose a root α .

4. Compute

$$m = \sqrt{2\alpha - p}.$$

5. Compute

$$n = \frac{q}{2m}.$$

6. Solve the quadratics

$$y^2 - my + \alpha + n = 0,$$

$$y^2 + my + \alpha - n = 0.$$

7. Recover

$$x = y - \frac{a}{4}.$$

8 Computational Complexity

Ferrari's method has finite algebraic complexity.

Operation	Complexity
Polynomial reduction	$O(1)$
Resolvent cubic solution	$O(1)$
Quadratic solutions	$O(1)$
Radical evaluations	finite

Table 2: Computational structure of Ferrari's method

The quartic is therefore recursively reducible into lower-degree solvable systems.

9 Geometry of Quartic Functions

Consider the quartic function

$$f(x) = x^4 - 3x^2 + 1.$$

Its derivative is

$$f'(x) = 4x^3 - 6x.$$

Critical points satisfy

$$2x(2x^2 - 3) = 0.$$

Thus:

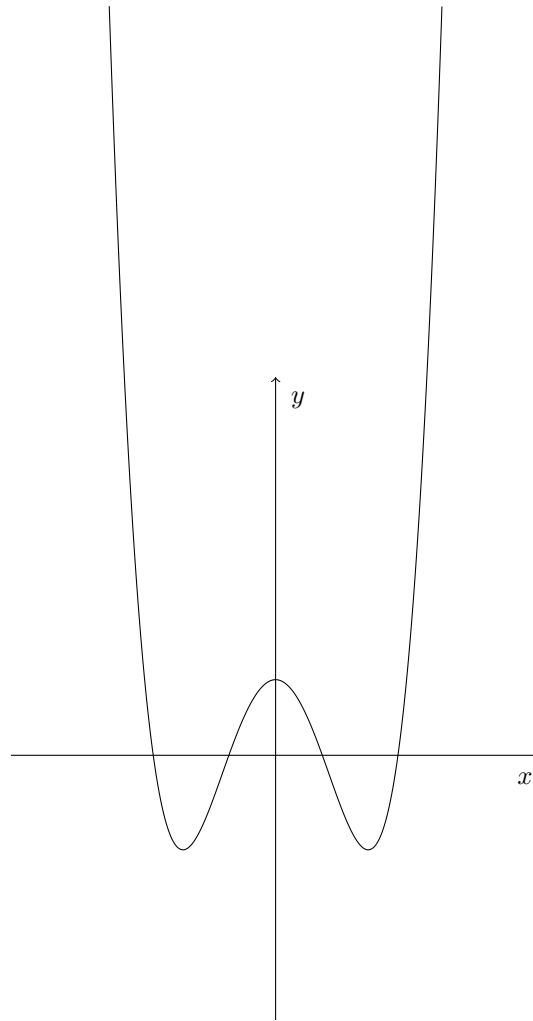
$$x = 0,$$

or

$$x = \pm\sqrt{\frac{3}{2}}.$$

Quartic functions therefore admit rich geometric structures involving multiple extrema and bifurcation-like transitions.

9.1 Quartic Geometry Illustration



The quartic naturally generates:

- multiple equilibria,
- nonlinear instability,
- metastability,
- double-well structures.

10 Quartics in Economics and Finance

10.1 Economic Potential Functions

Nonlinear economic systems often exhibit quartic structure.

Consider

$$U(x) = x^4 - \alpha x^2 + \beta x.$$

Such functions generate:

- multiple local equilibria,
- nonlinear incentive structures,
- regime transitions,
- strategic hysteresis.

10.2 Financial Instability

Quartic corrections arise in:

- stochastic volatility systems,
- nonlinear derivative pricing,
- tail-risk analysis,
- systemic contagion models.

Fourth moments also define kurtosis:

$$\kappa = \frac{E[(X - \mu)^4]}{\sigma^4}.$$

Thus quartic structure governs extreme-event sensitivity.

11 Political Science and Geopolitics

Political systems frequently exhibit nonlinear equilibrium structures.

Suppose state stability is modeled by

$$S(x) = x^4 - \alpha x^2 + \beta x + \gamma.$$

Then:

- local minima correspond to stable regimes,
- maxima correspond to unstable political thresholds,
- bifurcations correspond to revolutionary transitions.

Quartic interaction structures emerge naturally in:

- alliance systems,
- deterrence equilibria,
- arms-race dynamics,
- economic warfare,
- strategic logistics.

12 Artificial Intelligence and Optimization

Quartic approximations arise naturally in machine learning optimization landscapes.

Near equilibrium,

$$L(\theta) \approx a\theta^4 + b\theta^3 + c\theta^2 + d\theta + e.$$

Quartic structure governs:

- saddle geometry,
- metastable minima,
- optimization curvature,
- nonlinear training dynamics.

Quartic potentials also appear in:

- Hopfield networks,
- energy-based models,
- nonlinear recurrent systems,
- statistical learning theory.

13 Computer Science and Algorithms

Ferrari's method demonstrates several foundational computational principles:

1. symbolic preprocessing,
2. recursive decomposition,
3. auxiliary-variable construction,
4. algebraic factorization.

These ideas anticipate modern symbolic computation systems and automated algebra engines.

13.1 Pseudo-Code

INPUT: a, b, c, d

Compute:

$$p = b - 3a^2/8$$

$$q = c - ab/2 + a^3/8$$

$$r = d - ac/4 + a^2b/16 - 3a^4/256$$

Solve:

$$2u^3 - pu^2 - 2ru + pr - q^2/4 = 0$$

Choose root u

$$m = \sqrt{2u - p}$$

$$n = q/(2m)$$

Solve:

$$y^2 - my + u + n = 0$$

$$y^2 + my + u - n = 0$$

Recover:

$$x = y - a/4$$

14 Philosophical Interpretation

The quartic represents a profound epistemological threshold.

Ferrari's method demonstrates that apparent complexity may conceal latent reducible structure.

The quartic therefore embodies several universal principles:

- recursive solvability,
- hidden symmetry,
- structural decomposition,
- algebraic reductionism.

These principles extend far beyond mathematics into economics, geopolitics, computation, and scientific reasoning itself.

15 Conclusion

The quartic equation is among the most consequential structures in the history of mathematics.

It represents:

- the culmination of classical radical algebra,
- a universal nonlinear equilibrium structure,
- and a prototype for recursive computational reduction.

Ferrari's method remains one of the great achievements of symbolic civilization, demonstrating that sufficiently deep structural insight can transform apparently intractable complexity into recursively solvable form.

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Glossary

Quartic

A polynomial equation of degree four.

Depressed Quartic

A quartic polynomial with vanishing cubic term.

Resolvent Cubic

Auxiliary cubic equation arising in Ferrari's method.

Radical Solvability

Exact solvability using arithmetic operations and radicals.

Discriminant

Polynomial invariant governing root multiplicity.

Bifurcation

Abrupt structural transition in a nonlinear system.

Kurtosis

Normalized fourth statistical moment.

Metastability

Temporary stability near a local equilibrium.

Double-Well Potential

Quartic energy landscape with two local minima.

The End